

Comparative Statics Analysis of a Fully Closed City

DAVID PINES AND EFRAIM SADKA*

*Foerder Institute for Economic Research, Faculty of Social Sciences,
Tel-Aviv University, Ramat Aviv, Israel*

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This paper extends the existing comparative statics analysis about open or semiclosed cities, notably that of W. Wheaton [*J. Econom. Theor.*, 9 (2), 223-237 (1974)], to a fully closed city, where urban land rent is redistributed to the city population. Specifically, we examine the effects on changes in exogenous variables, such as population size, agricultural rent, and initial endowments, on endogenous variables such as welfare level, city size, and rent schedule. Most, but not all, of the existing results are reestablished. Surprisingly, it is shown that, in contrast to the existing literature, and perhaps to intuition, the city area may shrink when the city population increases. The paper also extends the existing literature by analyzing the effect of the exogenous variables on the endogenous income and the demand for transportation. It is shown that, under plausible assumptions, the quantity of transportation demanded declines with its price. © 1986 Academic Press, Inc.

1. INTRODUCTION

In this paper we extend the literature on comparative statics analysis of the determinants of intraurban structure in two respects. First, we apply it to a fully closed city, where the rent is redistributed to the urban residents rather than being paid to absentee landlords. Second, we examine, among other things, the effects of the exogenous variables on per capita income, and the aggregate demand for transportation.

With a few exceptions the comparative statics analysis in the existing literature is confined to either fully open or semiclosed cities (e.g., Pines [4] or Wheaton [7]). An open city is defined as one with an exogenous utility level and endogenous population size; a semiclosed city is one with an endogenous utility level and exogenous population size. But in both definitions the income level is exogenous which means that landlords are absentee. Equivalently, the exogeneity of income implies that the nonland resources consumed locally are equal to production less total urban land rent.

In the fully closed city which is discussed in this paper, urban land is rented from absentee landlords at a price equaling the *agricultural* (rather

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than the urban) rent. Thus, each city resident's income must include a share in the total urban rent above the agricultural rent, or the differential land rent (DLR), using the term of Arnott and Stiglitz [1]. Thus, the resident's income becomes endogenous.

The importance of the distinction between comparative statics analysis for a fully closed and a semiclosed city was discussed by Solow [6] and Oron *et al.* [3], and was also recognized by Wheaton [8]. Not all the comparative statics analysis of a semiclosed city can be carried through to the case of a fully closed city. Furthermore, some policy implications which apparently follow from a comparative statics analysis of a semiclosed city may be misleading, as will be illustrated later.

Although being recognized as important theoretical issues, the efforts to extend the analysis to a fully closed city were so far confined to specific functions and numerical calculations (see, e.g., Solow [6] and Robson and Scheffman [5]). In this paper we use analytical techniques to study the comparative statics properties of a fully closed city. We extend Wheaton's results and derive some new ones. In Section 2 we present the model and its basic assumptions. Section 3 is devoted to the comparative statics analysis. First, we analyze how the welfare level is affected by the exogenous parameters. The latter include the population size, the initial endowment, the per-mile transportation cost per person, and the agricultural rent or the opportunity cost of land. Secondly, we examine the effect of each exogenous parameter on the urban rent function, the city's boundary, total demand for transportation, and total income. In Section 4 we summarize the results and discuss some of their implications.

2. A MODEL OF A FULLY CLOSED CITY

Consider a monocentric city such that location is defined by distance from the city center x . Each of the N households living in the city owns a certain quantity I of a composite good. The city's land is rented from agricultural landlords at a fixed price R_A , which reflects the marginal productivity of land in its agricultural use. This land is rented out to urban use at a price $R(x)$ per unit which varies with the location of the land. Each resident receives an equal share $(1/N)$ of the profits (DLR) resulting from the difference between $R(x)$ and R_A . The usable land within a radius x is $\phi(x)$ and therefore total profits from land transactions are

$$\text{DLR} \stackrel{\text{def}}{=} \int_0^{\bar{x}} \dot{\phi}(x) (R(x) - R_A) dx$$

where $\dot{\theta}(x) = \dot{\phi}(x)$, $\bar{x}$ is the city's outer boundary, and a dot denotes derivative with respect to distance. Hence, the income Y of a representative

household is given by

$$Y = I + \frac{1}{N} \int_0^{\bar{x}} \theta(x) (R(x) - R_A) dx = I + \frac{1}{N} (\text{DLR}). \quad (1)$$

Households are assumed to have identical preferences represented by a quasiconcave utility function $u(Z, H)$ where Z is the quantity of the composite good and H is land used for housing. The price of the composite good is normalized to unity. We denote by $z(R(x), u)$ and $h(R(x), u)$ the compensated (Hicksian) demands for the composite good and land for housing, respectively, and by $E(R(x), u)$ the minimum expenditure of a person residing at x . (Notice that since the households have the same initial endowments, they must enjoy the same utility level u irrespective of x .) The income Y given by (1) must suffice for the minimum expenditure on Z and H and for transportation, i.e.,

$$E(R(x), u) + \alpha t(x) = Y \quad \text{for all } x, \quad (2)$$

where $\alpha t(x)$ is the cost of transportation from x to the city center. It is assumed the $t(0) = 0$, and $i > 0$.

Equality between the supply of and the demand for land in each location is represented by

$$n(x)h(R(x), u) - \theta(x) = 0, \quad (3)$$

where $n(x)$ is the number of households residing at x . Accommodating the city's population within the city's boundaries implies

$$N - \int_0^{\bar{x}} n(x) dx = 0; \quad (4)$$

$i > 0$ and (2) imply that $\dot{R} < 0$. Therefore, competition at the boundary $\bar{x}$ implies

$$R(\bar{x}) = R_A. \quad (5)$$

Equality between the supply of and the demand for the composite good within the city is represented by

$$\int_0^{\bar{x}} [n(x)(z(R(x), u) + \alpha t(x)) + \theta(x)R_A] dx = NI. \quad (6)$$

Notice that this last relation can be derived from (1)–(5) (Walras' law) and is therefore redundant. Alternatively, (1) can be viewed as redundant since it follows from (2)–(6).

In contrast to a semiclosed city model, in a fully closed one $(DLR)/N$ is included on the rhs of (1), which, together with (2)–(5), implies (6). In a semiclosed city, income is composed of I only while DLR accrues to the absentee landlords. Consequently, $R(x)$ replaces R_A on the lhs of (6).

One can verify that the allocation represented by (1)–(5) [or (2)–(6)] is identical to the one resulting from a maximization of u subject to the technological constraints (3), (4), and (6) (e.g., Oron *et al.* [3]). This result is not, of course, surprising, in view of the fact that there are no externalities so that the well-known association between competitive equilibria and Pareto-optima holds.

3. COMPARATIVE STATICS

In what follows we investigate the effects of marginal changes in the exogenous parameters I , α , R_A , and N on the endogenous variables u , $R(x)$, $\bar{x}$, Y , DLR, and T where $T = \int_0^{\bar{x}} n(x)t(x) dx$. T could be interpreted as the aggregate demand for transportation. For instance, when $t(x) = x$, then T is simply total miles driven. α could be interpreted as the cost per mile, so that $\alpha T = \int_0^{\bar{x}} n(x)\alpha t(x) dx$ is total transportation cost. Of course, when α is fixed, the aggregate demand for and the total cost of transportation vary in the same directions.

3.1. Effect of Exogenous Parameters on Utility

We begin by determining the relation between the resident's common welfare level u and the exogenous parameters. A total differentiation of (1) and (2), employing (5), yields, after some rearrangements,

$$\begin{aligned} dR(x) &= \frac{1}{E_R} [dY - t(x) d\alpha - E_u du] \\ &= \frac{1}{E_R} \left\{ dI - \frac{dN}{N^2} \int_0^{\bar{x}} \theta(x) (R(x) - R_A) dx \right. \\ &\quad \left. + \frac{1}{N} \int_0^{\bar{x}} \theta(x) (dR(x) - dR_A) dx - t(x) d\alpha - E_u du \right\}, \quad (7) \end{aligned}$$

where the arguments of the functions are suppressed when no ambiguity arises.

Multiply both sides by $n(x)$ and integrate with respect to x , using (3), (4), and the derivative property of the expenditure function (namely,

$E_R = h$) to get

$$du = \left[\int_0^{\bar{x}} n E_u dx \right]^{-1} \left\{ N dI - \frac{dN}{N} \int_0^{\bar{x}} \theta(x) (R(x) - R_A) dx \right. \\ \left. - d\alpha \int_0^{\bar{x}} n(x) t(x) dx - dR_A \int_0^{\bar{x}} \theta(x) dx \right\}. \quad (8)$$

Note that $E_u = 1/u_z > 0$ and $R(x) > R_A$ for all $x < \bar{x}$. Thus, we have

PROPOSITION 1. $\partial u / \partial I > 0$, $\partial u / \partial N < 0$, $\partial u / \partial \alpha < 0$, $\partial u / \partial R_A < 0$.

Note that one can alternatively derive Proposition 1 by applying the envelope theorem to the optimization model which is discussed at the end of the preceding section.

3.2. Some Preliminary Results and Assumptions

In determining the effects of the exogenous parameters on the other variables we follow Wheaton [7] with several short cuts.

Substituting (3) into (4) yields

$$\int_0^{\bar{x}} \frac{\theta}{h} dx = N. \quad (9)$$

By differentiating (2) with respect to x we obtain

$$\dot{R} = - \frac{\alpha \dot{t}}{h}. \quad (10)$$

Substituting (10) into (9) gives

$$- \int_0^{\bar{x}} \epsilon \dot{R} dx = \alpha N, \quad (11)$$

where $\epsilon \stackrel{\text{def}}{=} \theta / \dot{t}$. Integrate the lhs of (11) by parts, use (5), and assume that $\theta(0) = 0$ to obtain

$$\int_0^{\bar{x}} \dot{\epsilon} R dx - \epsilon(\bar{x}) R_A = \alpha N. \quad (12)$$

Total differentiation of (12) with respect to the exogenous parameters yields

$$\int_0^{\bar{x}} \dot{\epsilon} dR(x) dx = \epsilon(\bar{x}) dR_A + \alpha dN + N d\alpha. \quad (13)$$

To proceed at this stage we have to make the following assumptions.

Assumption 1. Land for housing (h) is a normal good, i.e., $h_u > 0$.

Assumption 2. $\dot{\epsilon} > 0$.

We adopt these two assumptions from now on. For some of the results we need to make an additional assumption. This assumption will be specifically mentioned whenever we make use of it.

Assumption 3. $d(\theta/\dot{\epsilon})/dx > 0$.

Assumptions 2 and 3 are satisfied, for example, in a circular city ($\theta(x) = 2\pi x$) with a linear transportation cost function ($\alpha t(x) = bx$).

3.3. Effects of a Change in I

Set $d\alpha = dR_A = dN = 0$ and consider the effect of a change in I . Equation (13) implies that

$$\int_0^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial I} dx = 0. \quad (13a)$$

In view of Assumption 2, $\partial R/\partial I$ must change sign at least once. Suppose then that x^1 is a point where $\partial R/\partial I$ changes sign, i.e., $\partial R(x^1)/\partial I = 0$. Differentiating (10) with respect to I implies that

$$\frac{\partial \dot{R}}{\partial I} = \frac{\alpha i}{h^2} \left(h_R \frac{\partial R}{\partial I} + h_u \frac{\partial u}{\partial I} \right) \Big|_{x=x^1} = \frac{\alpha i}{h^2} h_u \frac{\partial u}{\partial I} > 0,$$

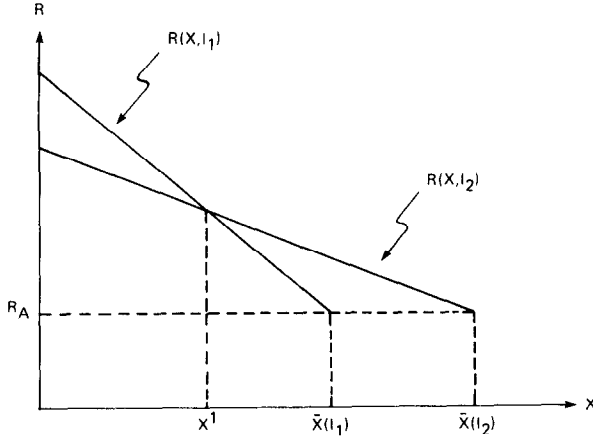
by the normality of h and Proposition 1. Hence, $\partial R/\partial I$ changes sign exactly once, being first negative and then positive.

Thus, Wheaton's [7] result regarding the effect of income on rent at the center and the boundary of the city is valid for a fully closed city as well. This is reflected in Fig. 1.

It follows from Fig. 1 that y increases with I . To see this, consider location x^1 , where $R(x^1)$ and $\alpha t(x^1)$ do not change when I increases. We know from Proposition 1 that u increases. When the rent and transportation cost are constant (at x^1) utility can increase only if income Y increases.

There still remains a question whether Y increases by more than I or less, i.e., whether DLR increases or decreases. It turns out that this depends on whether Assumption 3 is satisfied or not. Using (13a) we have

$$\frac{\partial \text{DLR}}{\partial I} = \int_0^{\bar{x}} \theta \frac{\partial R}{\partial I} + \theta(\bar{x})(R(\bar{x}) - R_A) \frac{\partial \bar{x}}{\partial I} = \int_0^{\bar{x}} \theta \frac{\partial R}{\partial I} = \int_0^{\bar{x}} \frac{\theta}{\dot{\epsilon}} \dot{\epsilon} \frac{\partial R}{\partial I}.$$

FIG. 1. $I_1 < I_2$.

Since $\frac{\partial R}{\partial I}$ changes sign once, being first negative and then positive, and since $\int_0^{\bar{x}} \frac{\partial R}{\partial I} = 0$ [see (13a)], it follows that

$$\int_0^{\bar{x}} \frac{\theta}{\varepsilon} \frac{\partial R}{\partial I} \approx 0 \quad \Leftrightarrow \quad \frac{d(\theta/\varepsilon)}{dx} \approx 0.$$

Hence, with Assumption 3, $\partial \text{DLR} / \partial I > 0$.

It is slightly more difficult to determine the effect of I on the aggregate demand for transportation T . When I increases the density must fall (i.e., housing consumption $h(R(x), u)$ increases) within the circle of radius x^1 , since both utility increases and the rent declines. Then, people move from each location within the radius x^1 to some location beyond x^1 , but housing consumption need not increase beyond x^1 , since there the rent rises. We thus cannot show that the net migration of households in locations beyond x^1 is negative, as is the case for locations within the radius of x^1 . Yet we can show (Appendix A) that the number of households residing beyond any circle of radius $x \in (0, \bar{x}]$, i.e., $M(x) \stackrel{\text{def}}{=} \int_x^{\bar{x}} n(s) ds$ increases with I . This means that the number of people crossing a cordon line of radius $x \in (0, \bar{x}]$ increases with I . Then, we have (by integration by parts and using $M(\bar{x}) = \iota(0) = 0$ and $\dot{M} = -n$)

$$T = \int_0^{\bar{x}} nt = \int_0^{\bar{x}} Mt.$$

Since M increases everywhere with I , so does T itself. (A formal proof appears in Appendix A.)

We can summarize our results in

PROPOSITION 2. (i) *There exists $x^1 \in (0, \bar{x})$ such that $\partial R/\partial I \gtrless 0$ as $x \gtrless x^1$; in particular, $\partial R(0)/\partial I < 0$;*

(ii) $\partial \bar{x}/\partial I > 0$;

(iii) $\partial Y/\partial I > 0$;

(iv) $\partial T/\partial I > 0$;

(v) *under Assumption 3; $\partial \text{DLR}/\partial I > 0$.*

3.4. Effects of a Change in α

Set $dI = dN = dR_A = 0$ in (13) to get

$$\int_0^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial \alpha} dx = N. \quad (13b)$$

Hence, $\partial R/\partial \alpha$ cannot be negative everywhere (recall that $\dot{\epsilon} > 0$, by Assumption 2). It is also true that $\partial R/\partial \alpha$ cannot be positive everywhere. To see this, observe that in view of Proposition 1, $h(R(x), u)$ must decline everywhere if R increases everywhere (since $h_u > 0$ and $h_R < 0$). Hence, the density must increase everywhere and therefore more people than N are located within the city, i.e., (4) is violated.¹ We therefore conclude that $\partial R/\partial \alpha$ changes sign. We can now use similar arguments as used for the effect of I to show that R changes sign only once. More specifically, we can show, as does Wheaton [7] for the case of a semiclosed city, that in our case too, the rent increases close to the center and declines close to the periphery when α rises (see Fig. 2). This also implies that $\partial \bar{x}/\partial \alpha < 0$.

The effect of α on T has been studied by Robson and Scheffman [5], but they do not derive a general result and had to resort to numerical calculations. Using similar considerations as in the case of a change in I , we can show that T declines with α . According to Fig. 2 and Proposition 1, it is clear that the density increases for $x \in [0, x^2]$, but this need not be the case

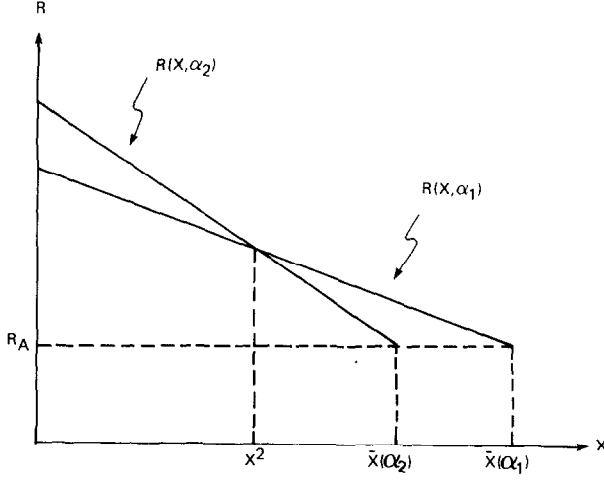
¹Formally, suppose that $\partial R/\partial \alpha \geq 0$ everywhere. It follows from (9) that

$$-\int_0^{\bar{x}} \frac{\theta}{h^2} \left(h_R \frac{\partial R}{\partial \alpha} + h_u \frac{\partial u}{\partial \alpha} \right) dx + \frac{\theta(\bar{x})}{h(R_A, u)} \frac{\partial \bar{x}}{\partial \alpha} = 0,$$

which implies that $\partial \bar{x}/\partial \alpha < 0$. But it follows from (5) that

$$\dot{R}(\bar{x}) \frac{\partial \bar{x}}{\partial \alpha} + \frac{\partial R(\bar{x})}{\partial \alpha} = 0,$$

which is a contradiction, because $\dot{R} < 0$.

FIG. 2. $\alpha_1 < \alpha_2$.

for $x > x^2$. Yet we show in Appendix A that $M(x) = \int_x^{\bar{x}} n(s) ds$ declines for every $x \in (0, \bar{x})$, which implies that T declines with α .

As was noted earlier, α can be interpreted as the price and T as the quantity of transportation demanded. Thus, the above relation implies that the quantity of transportation demanded declines with its price. It will also be interesting to study the elasticity of the demand for transportation (T) with respect to the unit cost of transportation (α). The magnitude of this elasticity (i.e., whether higher or lower than unity) determines the effect of α on total transportation cost (αT). This is a quite difficult problem which is studied elsewhere by Arnott, *et al.* [2].

These results are summarized in

PROPOSITION 3. (i) *There exists $x^2 \in (0, \bar{x})$ such that $\partial R / \partial \alpha \geq 0$ as $x \leq x^2$. In particular, $\partial R(0) / \partial \alpha > 0$;*

(ii) $\partial \bar{x} / \partial \alpha < 0$;

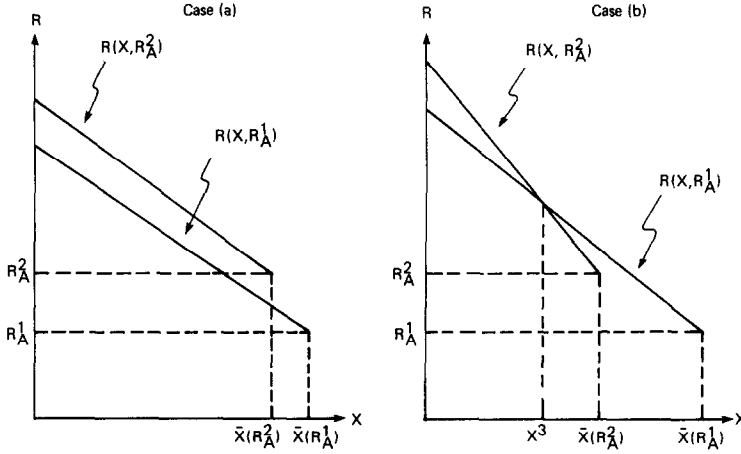
(iii) $\partial T / \partial \alpha < 0$.

3.5. Effects of a Change in R_A

Setting $d\alpha = dI = dN = 0$ in (13) we obtain

$$\int_0^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial R_A} dx = \epsilon(\bar{x}). \quad (13c)$$

Hence, $\partial R / \partial R_A$ must be positive somewhere. Therefore, if $\partial R(0) / \partial R_A \leq$

FIG. 3. $R_A^1 < R_A^2$.

0, then there must exist $x' \in (0, \bar{x})$ such that $\partial R(x')/\partial R_A = 0$, and $\partial \dot{R}(x')/\partial R_A \geq 0$. But since, by (10),

$$\frac{\partial \dot{R}}{\partial R_A} = \frac{\alpha i}{h^2} \left(h_R \frac{\partial R}{\partial R_A} + h_u \frac{\partial u}{\partial R_A} \right)_{x=x'} = \frac{\alpha i}{h^2} h_u \frac{\partial u}{\partial R_A} < 0,$$

we have a contradiction. Thus, we conclude that $\partial R(0)/\partial R_A > 0$. But we cannot determine the sign of $\partial R/\partial R_A$ at locations other than the city center. However, only two possibilities can occur: either $\partial R/\partial R_A$ is positive everywhere or, as before, we can show that it changes sign only once, being first positive and then negative. In Appendix B we present an example which shows that these two possibilities can, in fact, occur.

If $\partial R/\partial R_A$ changes sign, it is certainly true that $\partial \bar{x}/\partial R_A < 0$. But this must be true even when $\partial R/\partial R_A \geq 0$ everywhere, since the density increases everywhere because h declines due to the decrease in u which is implied by Proposition 1 and the substitution effect implied by the increase in R (see Fig. 3)

T must decrease with R_A . If R increases everywhere this is straightforward since the density increases everywhere. But this is also true if R changes sign (see Appendix A).

When $\partial R/\partial R_A$ changes sign it must be the case that $\partial Y/\partial R_A < 0$. The reason is that in this case there exists x^3 such that $\partial R(x^3)/\partial R_A = 0$. But by Proposition 1 the utility declines. This can happen only if income decreases, since the price $R(x^3)$ remains the same. In the case where $R(x)$

increases everywhere we show in Appendix A that

$$\partial \text{DLR} / \partial R_A = \partial Y / \partial R_A \leq 0 \quad \Leftrightarrow \quad d(\theta / \dot{\epsilon}) dx \geq 0.$$

Hence, if Assumption 3 holds, then $\partial \text{DLR} / \partial R_A < 0$.

PROPOSITION 4. (i) *Either (a) $\partial R / \partial R_A > 0$ for all $x \in [0, \bar{x}]$, or (b) there exists $x^3 \in (0, \bar{x})$ such that $\partial R / \partial R_A \leq 0$ as $x \leq x^3$. In particular, $\partial R(0) / \partial R_A > 0$;*

$$(ii) \quad \partial \bar{x} / \partial R_A < 0;$$

$$(iii) \quad \partial T / \partial R_A < 0;$$

(iv) *In case (a), we have $\partial \text{DLR} / \partial R_A = \partial Y / \partial R_A < 0$, if Assumption 3 holds; In case (b), we have $\partial \text{DLR} / \partial R_A = \partial Y / \partial R_A < 0$.*

3.6. Effects of a Change in N

Setting $dI = dx = dR_A = 0$ in (13), we obtain

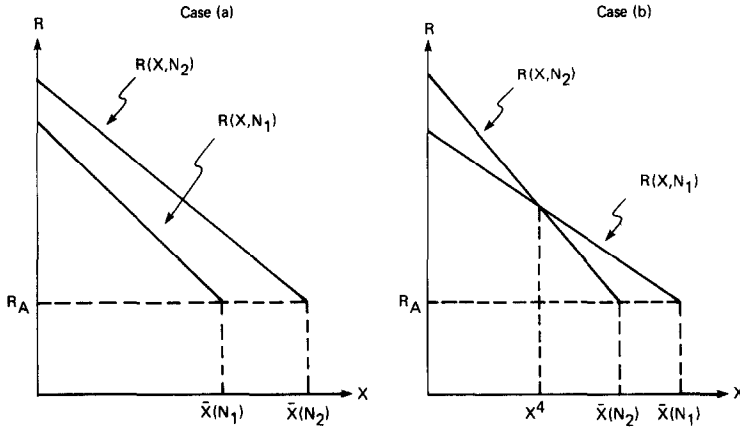
$$\int_0^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial N} dx = \alpha. \quad (13d)$$

Hence $\partial R / \partial N$ cannot be nonpositive everywhere. Therefore, if $\partial R(0) / \partial N \leq 0$, then there must exist $x'' \in [0, \bar{x})$ such that $\partial R(x'') / \partial N = 0$ and $\partial \dot{R}(x'') / \partial N \geq 0$. But since, by (10),

$$\frac{\partial \dot{R}}{\partial N} = \frac{\alpha \dot{t}}{h^2} \left(h_R \frac{\partial R}{\partial N} + h_u \frac{\partial u}{\partial N} \right) \Big|_{x=x''} = \frac{\alpha \dot{t}}{h^2} h_u \frac{\partial u}{\partial N} < 0,$$

we have a contradiction. Thus, we conclude that $\partial R(0) / \partial N > 0$. Since we have also shown that $\partial \dot{R} / \partial N < 0$ whenever $\partial R / \partial N = 0$, then either (a) $\partial R / \partial N > 0$ for all $x \in [0, \bar{x}]$ or (b) there exists x^4 such that $\partial R / \partial N \leq 0$ for $x \leq x^4$. Since, by (5), $\dot{R}(\bar{x}) \partial \bar{x} / \partial N + \partial R(\bar{x}) / \partial N = 0$, it follows that in case (a) we have $\partial \bar{x} / \partial N > 0$, while in case (b) we have $\partial \bar{x} / \partial N < 0$ (see Fig. 4).

Note that case (b) seems counterintuitive: an increase in the population lowers the rent on the fixed resource (land) in some locations of the city and shrinks the city's size. An explanation for this counterintuitive result may be provided by the income effect generated by a change in N . As has been shown in Section 3.1, an increase in N lowers u . Thus, an increase in N works like a reduction in real income I . As has been shown in Section 3.3, a decrease in I lowers the rent close to the city's boundary and reduces $\bar{x}$.

FIG. 4. $N_1 < N_2$.

Therefore, one will expect case (b) to occur when the income effect on housing is very strong: an increase in N lowers welfare and sharply reduces the demand of each household for residential land; consequently, total demand for residential land falls and $\bar{x}$ shrinks.² In Appendix C we provide an example where this occurs.

The effect of a change in N on Y or on T is ambiguous. We can only say that in case (b) Y must fall, while in case (a) T must rise. To see this, observe that in case (b) we have a location x^4 , where the rent does not change but u falls. This can happen if and only if income Y falls.³ In case (a) we have h falling everywhere. Therefore more people reside at every location and total transportation demand is higher.⁴

²This cannot happen whenever the utility is Cobb–Douglas and the Arnott–Stiglitz condition for constant ATC/DLR ratio is satisfied (see Arnott and Stiglitz [1]).

³Formally, it follows from (7) that

$$\frac{\partial R(x^4)}{\partial N} = \frac{1}{h(R(x^4), u)} \left[\frac{\partial Y}{\partial N} - E_u(R(x^4), u) \frac{\partial u}{\partial N} \right].$$

Since $\partial R(x^4)/\partial N = 0$ and $\partial u/\partial N < 0$, it follows that $\partial Y/\partial N < 0$.

⁴Formally,

$$\frac{\partial T}{\partial N} = \frac{\partial}{\partial N} \int_0^{\bar{x}} nt = \int_0^{\bar{x}} \frac{\partial n}{\partial N} t dx + n(\bar{x}) t(\bar{x}) \frac{\partial \bar{x}}{\partial N}.$$

Since $\partial \bar{x}/\partial N > 0$ and since $\partial n/\partial N = \partial(\theta/h)/\partial N = -\theta h^{-2}(h_R \partial R/\partial N + h_u \partial u/\partial N) > 0$, it follows that $\partial T/\partial N > 0$.

The results regarding the effect of N are summarized in

PROPOSITION 5. (i) *Either (a) $\partial R/\partial N > 0$ for all $x \in [0, \bar{x})$ or (b) there exists $x^4 \in (0, \bar{x})$ such that $\partial R/\partial N \gtrless 0$ as $x \lessgtr x^4$. In particular, $\partial R(0)/\partial N > 0$;*

(ii) *In case (a), $\partial \bar{x}/\partial N > 0$; in case (b) $\partial \bar{x}/\partial N < 0$;*

(iii) *In case (a), $\partial T/\partial N > 0$;*

(iv) *In case (b), $\partial Y/\partial N < 0$;*

(v) *In case (a), $\partial DLR/\partial N > 0$.*

4. SUMMARY AND CONCLUDING REMARKS

The results of the comparative statics are summarized in Table 1. The exogenous variables are listed at the top and the endogenous variables at the left. Recall that there are two possibilities, labeled (a) and (b), for the effects of changes in R_A and N .

The results in Table 1 can be compared to those of Wheaton [7]. There are two differences—the effects of R_A and N on urban land rent close to the city's boundary and the effect of N on the city's area are ambiguous in our closed city, contrary to Wheaton's city.

Although the results are almost identical and in both cases a competitive equilibrium without externalities is assumed, the implications may be different, since the distribution of the initial endowments in the two models are different. In order to illustrate this difference, consider the following statement which is often made: "There is no reason to contain suburbanization resulting from an increase in productivity, since this process is consistent with a competitive equilibrium." As shown in Section 3.2 an increase in I is associated not only with suburbanization but also with an increase in the DLR. In a closed city, where the DLR is part of the residents' income,

TABLE 1

	I	α	R_A		N	
			(a)	(b)	(a)	(b)
u	+	—	—	—	—	—
$R(x)$						
$x = 0$	—	+	+	+	+	+
$x = \bar{x}$	+	—	+	—	+	—
$\bar{x}$	+	—	—	—	+	—
T	+	—	—	—	+	?
Y	+	?	— ^a	—	?	—
DLR	+ ^a	?	— ^a	—	+	?

^aIf Assumption 3 is satisfied.

the equilibrium allocation maximizes the (equal) utility level of the city population. But in a semiclosed city, discussed, for example, by Wheaton, the DLR leaks out to the absentee landlords. It is in the interests of the city population to reduce this leakage. Therefore, it may be advisable from the point of view of the city's population to contain suburbanization. In order to see that this may be advantageous, we use the result that limiting the city's size by, say, an artificial increase in R_A may reduce the DLR. As long as the disadvantage of the inefficiency resulting from containing the suburbanization is more than being offset by the advantage of more favorable distribution of income, the policy may be advisable. This example illustrates the importance of the distinction between the two models and the need for comparative statics for a fully closed city.

APPENDIX A

Proof of Proposition 2

Part (i) has already been proved in the text. In order to prove part (ii) notice that (5) implies that $\partial R(\bar{x})/\partial I + \dot{R}(\bar{x}) \partial \bar{x}/\partial I = 0$. Since $\partial R(\bar{x})/\partial I > 0$ [from part (i)] and $\dot{R}(\bar{x}) < 0$, we must have $\partial \bar{x}/\partial I > 0$. To prove (iii), observe that since $\partial R(x^1)/\partial I = 0$, we have from (7)

$$0 = \frac{\partial Y}{\partial I} - E_u(R(x^1), u) \frac{\partial u}{\partial I}.$$

Since $\partial u/\partial I > 0$ and $E_u > 0$, it follows that $\partial Y/\partial I > 0$.

Part (iv) will be proved in three steps:

(a) $\partial n(x)/\partial I < 0$ for all $x \in [0, x^1]$. Employing (3) we can see that

$$\frac{\partial n(x)}{\partial I} = -\frac{\theta}{h^2} \left(h_R \frac{\partial R}{\partial I} + h_u \frac{\partial u}{\partial I} \right) < 0,$$

because $\partial R/\partial I < 0$ [part (i)], $h_R < 0$, $h_u > 0$ (normality of h), and $\partial u/\partial I > 0$ (Proposition 1).

(b) $\partial M(x)/\partial I > 0$ for $x \in [0, \bar{x}]$. Part (a) implies that (b) holds for $x \in [0, x^1]$. If (b) does not hold for all x , then, by continuity considerations, there must exist $x^* \in (x^1, \bar{x}]$ such that

$$\frac{\partial M(x^*)}{\partial I} = - \int_0^{x^*} \frac{\partial n(x)}{\partial I} dx = 0. \quad (A1)$$

It follows from (13a) that

$$\begin{aligned} 0 &= \int_0^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial I} dx = \int_0^{x^*} \dot{\epsilon} \frac{\partial R}{\partial I} dx + \int_{x^*}^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial I} dx \\ &= \epsilon(x^*) \frac{\partial R(x^*)}{\partial I} - \int_0^{x^*} \epsilon \frac{\partial \dot{R}}{\partial I} dx + \int_{x^*}^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial I} dx, \end{aligned} \quad (A2)$$

where use is made of $\varepsilon(0) = \theta(0)/i(0) = 0$. Since $\dot{R} = -\alpha i/h = -\alpha i n/\theta = -\alpha n/\varepsilon$, it follows that

$$\varepsilon \partial \dot{R} / \partial I = -\alpha \partial n / \partial I. \quad (\text{A3})$$

Substituting (A3) into (A2) and rearranging yields

$$\int_{x^*}^{\bar{x}} \dot{\varepsilon} \frac{\partial R}{\partial I} dx + \varepsilon(x^*) \frac{\partial R(x^*)}{\partial I} = -\alpha \int_0^{x^*} \frac{\partial n}{\partial I} dx = 0, \quad (\text{A4})$$

by (A1). But this is a contradiction, since $x^* > x^1$ and, by part (i), $\partial R / \partial I > 0$ for $x \geq x^1$.

(c) $\partial T / \partial I > 0$. By integration by parts [and recalling that $M(\bar{x}) = 0 = t(0)$],

$$\frac{\partial T}{\partial I} = \int_0^{\bar{x}} \frac{\partial M(x)}{\partial I} i(x) dx + M(\bar{x}) i(\bar{x}) \frac{\partial \bar{x}}{\partial I} > 0,$$

by (ii) and part (b).

Q.E.D.

Proof of Proposition 3(iii)

The proof is analogous to that of Proposition 2(iv).

Step (a) now becomes $\partial n(x) / \partial \alpha > 0$ for all $x \in [0, x^2]$. Its proof follows immediately by replacing ∂I by $\partial \alpha$. Step (b) now becomes $\partial M(x) / \partial \alpha < 0$ for all $x \in [0, \bar{x}]$. Its proof proceeds as in Proposition 2, except that (A2) becomes

$$N = \varepsilon(x^*) \frac{\partial R(x^*)}{\partial \alpha} - \int_0^{x^*} \varepsilon \frac{\partial \dot{R}}{\partial \alpha} dx + \int_{x^*}^{\bar{x}} \dot{\varepsilon} \frac{\partial R}{\partial \alpha} dx, \quad (\text{A2})'$$

(A3) becomes

$$\varepsilon \partial \dot{R} / \partial \alpha = -n - \alpha \partial n / \partial \alpha, \quad (\text{A3})'$$

and (A4) becomes

$$\begin{aligned} \int_{x^*}^{\bar{x}} \dot{\varepsilon} \frac{\partial R}{\partial \alpha} dx + \varepsilon(x^*) \frac{\partial R(x^*)}{\partial \alpha} &= N - \int_0^{x^*} n dx - \alpha \int_0^{x^*} \frac{\partial n}{\partial \alpha} dx \\ &= N - \int_0^{x^*} n dx > 0, \end{aligned} \quad (\text{A4})'$$

in contradiction to part (i) because $x^* > x^2$. Step (c) becomes $\partial T / \partial \alpha < 0$ and its proof is as before.

Q.E.D.

Proof of Proposition 4(iii)

In case (a) the density increases everywhere and the proof is straightforward. The problem is with case (b) where $\partial R/\partial R_A \leq 0$ when $x \geq x^3$. Step (a) in the proof of Proposition 2(iii) becomes $\partial n(x)/\partial R_A > 0$ for all $x \in [0, x^3]$. Its proof follows immediately by replacing ∂I by ∂R_A . Step (b) now becomes $\partial M(x)/\partial R_A < 0$ for all $x \in [0, \bar{x}]$. Its proof proceeds as in Proposition 2, except that (A2) becomes

$$\epsilon(\bar{x}) = \dot{\epsilon}(x^*) \frac{\partial R(x^*)}{\partial R_A} - \int_0^{x^*} \epsilon \frac{\partial \dot{R}}{\partial R_A} dx + \int_{x^*}^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial R_A}, \quad (\text{A2}'')$$

(A3) becomes

$$\epsilon \partial \dot{R}/\partial \alpha = -\alpha \partial n/\partial \alpha, \quad (\text{A3}'')$$

and (A4) becomes

$$\int_{x^*}^{\bar{x}} \dot{\epsilon} \frac{\partial R}{\partial R_A} dx + \epsilon(x^*) \frac{\partial R(x^*)}{\partial R_A} = \epsilon(\bar{x}), \quad (\text{A4}'')$$

since

$$\int_0^{x^*} \epsilon \frac{\partial \dot{R}}{\partial R_A} dx = - \int_0^{x^*} \alpha \frac{\partial n}{\partial R_A} dx = -\alpha \frac{\partial M(x^*)}{\partial R_A} = 0.$$

Now in case (b) (A4)'' implies a contradiction since $x^* > x^3$ and therefore $\partial R/\partial R_A < 0$ for all $x \geq x^*$, and since $\epsilon(\bar{x}) > 0$. Q.E.D.

Proof of Proposition 4(iv)

Consider first case (a). We observe from (1) that

$$\begin{aligned} \partial Y/\partial R_A &= \partial \left[\int_0^{\bar{x}} \theta(R - R_A) dx \right] / \partial R_A \\ &= \int_0^{\bar{x}} \theta \left(\frac{\partial R}{\partial R_A} - 1 \right) dx = \int_0^{\bar{x}} \frac{\theta}{\dot{\epsilon}} \epsilon \left(\frac{\partial R}{\partial R_A} - 1 \right) dx. \end{aligned} \quad (\text{A5})$$

Since $\epsilon(0) = \theta(0)/i(0) = 0$, it follows that

$$\epsilon(\bar{x}) = \int_0^{\bar{x}} \dot{\epsilon}(x) dx.$$

Hence, (13c) can be rewritten as

$$\int_0^{\bar{x}} \dot{\varepsilon} \left(\frac{\partial R}{\partial R_A} - 1 \right) dx = 0. \quad (\text{A6})$$

By differentiating (10) with respect to R_A , we get

$$\frac{\partial \dot{R}}{\partial R_A} = \frac{\alpha i}{h^2} \left(h_R \frac{\partial R}{\partial R_A} + h_u \frac{\partial u}{\partial R_A} \right) < 0. \quad (\text{A7})$$

Thus, it follows from (A6) and (A7) that $\partial R / \partial R_A - 1$ changes sign once, being first positive and then negative. Since $\dot{\varepsilon} > 0$, then $\dot{\varepsilon}(\partial R / \partial R_A - 1)$ changes sign once, being first positive and then negative. Hence, it follows from (A5) and (A6) that a sufficient condition for $\partial Y / \partial R_A$ to be negative is that $\theta / \dot{\varepsilon}$ be increasing in x .

Case (b) was already proved in the text

Q.E.D.

APPENDIX B

In the text we showed that $\partial R / \partial R_A$ can follow only two patterns: either it is positive everywhere, or it changes sign once, being first positive and then negative. Here we present an example to show that these two possibilities can actually occur.

Consider a Leontief utility function

$$u(Z, H) = \min(Z, H)$$

which gives rise to the following expenditure and compensated demand functions:

$$E(R, u) = (R + 1)u, \quad h(R, u) = u, \quad \text{and} \quad z(R, u) = u.$$

Suppose that the city is circular [$\theta(x) = 2\pi x$] and that the transportation cost function is linear [$\alpha t(x) = bx$]. From (3) we conclude that

$$n = 2\pi x / u. \quad (\text{B1})$$

By (4) we have

$$\int_0^{\bar{x}} 2\pi x / u = N$$

and hence

$$u = \pi \bar{x}^2 / N. \quad (\text{B2})$$

Substituting (B1) and (B2) into the resource constraint (6) yields

$$\pi(1 + R_A)\bar{x}^2 + \frac{2}{3}bN\bar{x} - NI = 0. \quad (B3)$$

By differentiating (B3) with respect R_A we obtain

$$\frac{\partial \bar{x}}{\partial R_A} = - \frac{\pi \bar{x}^2}{2\pi(1 + R_A)\bar{x} + \frac{2}{3}bN} < 0, \quad (B4)$$

in accordance with Proposition 4. Equation (2) which states that total spending is constant at all x implies that

$$E(R(x), u) + bx = E(R_A, u) + b\bar{x}.$$

Substituting $E = (R + 1)u = (R + 1)\pi\bar{x}^2/N$, it follows that

$$R(x) = R_A + \frac{bN}{\pi\bar{x}^2}(\bar{x} - x). \quad (B5)$$

Differentiating (B5) with respect to R_A yields

$$\frac{\partial R(x)}{\partial R_A} = 1 + \frac{bN}{\pi\bar{x}^4}[\bar{x}^2 - 2\bar{x}(\bar{x} - x)] \frac{\partial \bar{x}}{\partial R_A}$$

and substituting (B4) we obtain

$$\frac{\partial R(x)}{\partial R_A} = 1 - \frac{bN[\bar{x}^2 - 2\bar{x}(\bar{x} - x)]}{\bar{x}^2[2\pi(1 + R_A)\bar{x} + \frac{2}{3}bN]}.$$

At $\bar{x}$, the latter derivative becomes

$$\frac{\partial R(\bar{x})}{\partial R_A} = 1 - \frac{bN}{2\pi(1 + R_A)\bar{x} + \frac{2}{3}bN}. \quad (B6)$$

As can be seen from (B3), $\bar{x}$ depends on the exogenous variables, b , N , I , and R_A . However, it can also be seen from (B3) that for any variations in bN , we can keep $\bar{x}$ constant by appropriate variations in NI . Thus, letting bN go to infinity, while appropriately changing NI to keep $\bar{x}$ constant, we can make $bN/[2\pi(1 + R_A)\bar{x} + 2bN/3]$ approach $\frac{3}{2}$ and thus, by (B6), $\partial R(\bar{x})/\partial R_A$ approaches $-\frac{1}{2} < 0$. On the other hand, for a small value of bN , we can see that $\partial R(\bar{x})/\partial R_A$ approaches one which is positive.

This example has a zero elasticity of substitution between H and Z . The reader can verify that the two possibilities mentioned above can occur also

with an infinite elasticity (i.e., $u = Z + H$). This suggests that these possibilities can occur at any level of the elasticity of substitution.

APPENDIX C

In this appendix we produce an example where case (b) of Section 3.6 occurs. As has been pointed out in the text, the demand for housing has to have a very large income elasticity. Specifically, suppose that H captures all the income effect and thus Z is a neutral good:

$$u = H + 2Z^{1/2}. \quad (C1)$$

In this case it is straightforward to show that the expenditure and compensated demand functions have the forms

$$\begin{aligned} E(R, u) &= uR - R^2, \\ z(R, u) &= R^2, \\ h(R, u) &= u - 2R. \end{aligned}$$

We consider parameter values where $R < u/2$ in order to avoid corner solutions. We assume that $R_A = 0$, so that $E(R(\bar{x}), u) = 0$, and $at(x) = x$.

It follows from (4) that

$$x + E(R(x), u) = E(R(\bar{x}), u) + \bar{x}$$

and, upon substitution from (C2), we conclude that

$$R(x) = \left\{ u - [u^2 - 4(\bar{x} - x)]^{1/2} \right\} / 2. \quad (C3)$$

Substituting $n = \theta/h = \theta/(u - 2R)$ and (C3) into (4) and assuming for the sake of simplicity, that $\theta(x) = x$, we find, by integration by parts, that

$$N = \frac{1}{2}\bar{x}u - \frac{1}{12}u^3 + \frac{1}{12}(u^2 - 4\bar{x})^{3/2}. \quad (C4)$$

Substituting $n = \theta/h$, (C2), and (C3) into the resource constraint (6) yields, upon integration by parts,

$$NI = \left(\frac{1}{2}u^2 - \bar{x} \right) N + \frac{3}{4}u\bar{x}^2 - \frac{1}{3}u^3\bar{x} + \frac{1}{30}u^5 - \frac{1}{30}(u^2 - 4\bar{x})^{5/2}. \quad (C5)$$

Equations (C4) and (C5) are used in order to solve numerically for u and x for a range of values of N . The value of I that is taken is 20,000. Some of the simulation results are given in Table 2. The results clearly indicate that as N rises, $\bar{x}$ rises at first [case (a)] but then falls [case (b)]. Welfare (u) falls in N , as we already know from Proposition 1.

TABLE 2

N	U	$\bar{x}$
1	2074.1	213.9
100.000	1730.1	18,563.7
200.000	1686.7	25,893.8
300.000	1221.3	26,904.4
400.000	1123.6	29,739.8
450.000	1014.6	29,918.9
480.000	953.3	29,908.2
490.000	934.5	29,905.1
500.000	915.8	29,889.7
1000.000	484.2	29,592.6
1600.000	339.3	28,610.8

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